

PRODUCT BACKLOG

CeDInt-SEF: Smart Energy Forecasting

DataPack - CeDInt Smart Building

Document purpose

A complete, implementation-oriented Product Backlog for the DataPack solution, derived from the Business Case and the supplied CeDInt building/floor-plan material. The backlog translates the approved scope into PM²-compatible work packages, backlog items, acceptance criteria, dependencies, quality gates and delivery increments.

Project	CeDInt-SEF: Smart Energy Forecasting / DataPack
Business Case	Version 1.3 - September 2026
Delivery target	30 November 2026 - final presentation; 14 December 2026 - final handover
Delivery approach	PM ² governance + Scrum/Agile execution + CRISP-DM data-mining lifecycle
Project Manager / Scrum Master	Jules Charvet
Project Owner	JOM
Project Sponsor / Approving Authority	JOM
Document language	English
Status	subject to Product Owner / Sponsor validation

Prepared from the supplied Business Case and four CeDInt screenshots/floor-plan visuals.

1. Source Basis and Product Context

The Business Case defines DataPack as a predictive energy-consumption initiative for the highly instrumented CeDInt facility at UPM. The building is treated as a measurable system organised through a Building → Floor → Zone hierarchy. Historical IoT telemetry is supplied as Excel extracts; the project is explicitly limited to data processing and machine learning and excludes physical sensor work, live pipeline modification and real-time streaming.

Source-grounded scope boundary

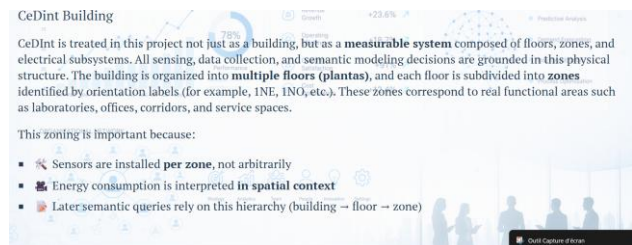
This backlog does not introduce hardware installation, live ThingsBoard/PostgreSQL integration, Node.js middleware changes or a real-time streaming architecture. Those are explicit Business Case exclusions. External data is treated as optional and only used if it improves analysis/model performance.

1.1 Building and spatial context captured in the supplied visuals

The supplied screenshots show CeDInt as a multi-floor building with floor plans, orientation-based zones and sensor categories represented on the plans. The Business Case and visuals support a spatial model in which sensors are associated with functional zones and zone identifiers encode floor and orientation (examples shown include 1NE and 1NO). The third-floor visual notes that the roof/third-floor plan has only four HVAC systems.



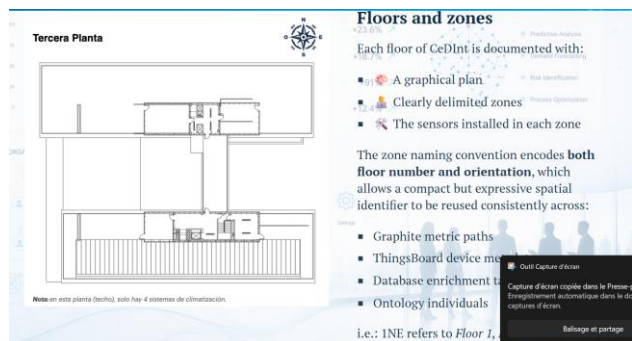
CeDInt location / site context



Floor 1 and Floor 2 sensor distribution



Third-floor plan and spatial naming context



Floor-plan / zone and sensor visual context

1.2 Business Case facts that directly drive the backlog

- Historical IoT data is provided through extracted Excel files for different stories/floors and zones.
- Known data issues include varying sampling frequencies and interference associated with unexpected device resets.
- The sensor/device families explicitly named in scope include energy meters BM321, BMT01 and BMP02; indoor environmental sensors BSens and BSA02; and smart-lighting nodes BSL01.
- BM321 devices may expose a large telemetry space (the Business Case gives approximately 90 telemetry keys per device).
- The spatial hierarchy is Building → Floor → Zone and existing unique device IDs are to be used for joining datasets.
- The predictive target is energy consumption; external context such as Madrid weather and the UPM/academic calendar may be integrated if useful.
- The methodology is CRISP-DM; execution is planned in two-week Agile/Scrum sprints.
- The hard presentation deadline is 30 November 2026; final code and documentation are due through Moodle on 14 December 2026.
- Financial cost is stated as €0, using open-source Python, Pandas and Scikit-learn plus free project-management/collaboration platforms.

2. PM² Methodology Alignment

PM² is used here as the project-governance framework, while Scrum supplies the iterative delivery mechanics and CRISP-DM structures the data-mining lifecycle. The Product Backlog is therefore the operational bridge between the approved Business Case and sprint-level delivery.

PM ² area	Relevant project concern	Backlog implementation
PM ² governance / initiation	Business Case, project objectives, scope, stakeholders, constraints	Backlog traceability to objectives, scope boundaries, sponsor acceptance and milestone dates
PM ² planning	Work Plan, schedule, resources, quality, risk and communications	Sprint mapping, estimates, dependencies, quality gates, DoR/DoD and release plan
PM ² execution	Deliverables and controlled project work	User stories/tasks implemented, reviewed and demonstrated at Sprint Reviews
PM ² monitoring & control	Progress, risks, issues, changes, decisions and quality	Backlog status, acceptance criteria, risk links, change-control rules and evidence
PM ² closing	Acceptance, handover, lessons learned and closure	Final model package, documentation, code, presentation evidence and Moodle handover

2.1 Product Backlog governance

- The backlog is owned at product level by the project/client side and validated at milestone/review points by the Sponsor / Approving Authority.
- The Project Manager / Scrum Master maintains backlog integrity, facilitates prioritisation and controls scope against the approved Business Case.
- Items are prioritised using MoSCoW plus delivery risk and dependency considerations; this is a planning mechanism, not a statement that one stakeholder interest is more important than another.
- Changes that affect approved scope, deadline, cost assumptions, exclusions or major deliverables should be raised through PM² change control rather than silently editing the baseline.
- Every backlog item has a testable acceptance condition and should produce evidence that can be inspected at Sprint Review or release.

2.2 Definition of Ready (DoR)

- Business outcome or technical purpose is explicit.
- Source data, device family, spatial entity or dependency is identified where relevant.
- Acceptance criteria are testable and understandable.
- Dependencies and assumptions are recorded.
- The item is small enough to complete within one sprint, or is explicitly treated as an epic/feature requiring decomposition.
- The item is consistent with the Business Case scope and does not introduce an excluded capability.

2.3 Definition of Done (DoD)

- Implementation or analysis is completed and reproducible from the project repository.
- Relevant data-quality checks pass and anomalies are documented rather than silently removed.
- Spatial mappings and joins are traceable to the central data dictionary.
- Code/notebooks/scripts are reviewed by the designated QA/model-validation member.
- Acceptance criteria are evidenced with test output, charts, tables, metrics or documentation as appropriate.
- No unresolved critical defect remains for the item.
- The result is linked to the applicable CRISP-DM phase and PM² project documentation.

3. Product Backlog Overview

The following backlog decomposes the approved DataPack scope into epics and user stories. Story points use a relative Fibonacci scale (1, 2, 3, 5, 8, 13). T-shirt estimates are deliberately omitted to avoid mixing estimation systems. Priority is MoSCoW: Must, Should, Could, Won't for this release.

Epic ID	Epic	Outcome	Release priority
EP-01	Project setup, governance & backlog control	Governed project foundation and controlled delivery	Must
EP-02	Data inventory & data dictionary	Authoritative inventory of files, telemetry, device IDs and spatial semantics	Must
EP-03	Data ingestion & quality engineering	Reliable ingestion and diagnosis of Excel historical data	Must
EP-04	Spatial modelling: building → floor → zone	Consistent spatial context for all analytical data	Must
EP-05	Feature engineering & contextual enrichment	Analysis-ready features from energy, environment, lighting and optional external context	Must
EP-06	Predictive modelling & evaluation	Baseline and improved energy-consumption forecasting models	Must
EP-07	Insights, visualisation & communication	Evidence-based findings and presentation material	Must
EP-08	Quality, risk, security & reproducibility	Cross-cutting controls and auditability	Must
EP-09	Final release, handover & closure	Complete delivery package and PM ² closure evidence	Must

3.1 Detailed Product Backlog

Total backlog items: 50 user stories, grouped into 9 epics. The stories below are the recommended implementation baseline; Sprint assignment remains adjustable through backlog refinement.

3.1 Detailed Product Backlog - Baseline

ID	Epic	Story / outcome	User story	Acceptance criteria	SP	MoSCoW	Dependencies	Target	CRISP-DM	Risk
US-001	EP-01	Establish project repository and branching convention	As a project team, we want a controlled repository so that code, notebooks, documentation and evidence are versioned and reproducible.	- Repository is created with agreed folders for data documentation, source code, notebooks, models, tests and deliverables. - A branch/merge convention is documented. - No raw restricted dataset is committed to version control unless explicitly authorised.	3	Must	None	S0	All	R4
US-002	EP-01	Baseline the Product Backlog	As the project team, we want an approved baseline backlog so that sprint commitments remain traceable to the Business Case.	- All epics are mapped to an approved scope area or delivery requirement. - Out-of-scope capabilities are explicitly flagged. - Priorities, dependencies and sprint targets are recorded.	3	Must	US-001	S0	Business Understanding	R4
US-003	EP-01	Set up sprint review and acceptance cadence	As stakeholders, we want regular reviews so that progress and scope remain controlled before the hard presentation deadline.	- Two-week sprint cadence is documented. - Each sprint has a review/demo and acceptance decision. - Unaccepted work is returned to the backlog with a reason.	2	Must	US-002	S0	Business Understanding	R4
US-004	EP-01	Maintain PM ² decision, change and risk traceability	As the project manager, I want decisions, changes and risks linked to backlog items so that governance evidence is auditable.	- Each material scope/schedule/technical decision has an owner and date. - Change requests identify impacted backlog items. - High risks have mitigation actions represented in backlog work.	3	Must	US-002	S0-S5	All	R4
US-005	EP-02	Inventory all supplied Excel extracts	As a data engineer, I want a complete inventory of source files so that ingestion is systematic and no supplied dataset is silently omitted.	- Each source file has a recorded filename, source period if available, floor/zone context if available, device family and expected content. - Duplicate or superseded extracts are identified. - Inventory is version-controlled.	3	Must	US-001	S1	Data Understanding	R1
US-006	EP-02	Catalogue telemetry by device family	As an analyst, I want telemetry keys catalogued by device family so that variables are interpreted consistently.	- BM321, BMT01, BMP02, BSens, BSA02 and BSL01 are represented where present in the extracts. - Units/meaning are documented when provided by the source. - Unknown keys are flagged rather than guessed.	5	Must	US-005	S1	Data Understanding	R2
US-007	EP-02	Create the central data dictionary	As the team, we want a single data dictionary so that device IDs, telemetry keys and spatial attributes are joined consistently.	- Unique device ID, device family, telemetry key, unit, source file, time field and spatial fields are represented. - Mapping provenance is recorded. - The dictionary is used by downstream transformations.	5	Must	US-006	S1	Data Understanding	R2
US-008	EP-02	Profile temporal coverage and granularity	As a data analyst, I want to understand temporal coverage and sampling frequency so that modelling windows are realistic.	- Start/end timestamps and observed sampling intervals are reported by relevant dataset/device family. - Irregular sampling is quantified. - Periods with insufficient coverage are flagged.	5	Must	US-005	S1	Data Understanding	R1
US-009	EP-02	Profile missingness and anomalies	As a data analyst, I want missingness and anomalous values quantified so that cleaning decisions are evidence-based.	- Missing values are reported by key/device/time period. - Candidate outliers and reset-related patterns are summarised. - Cleaning decisions are logged with rationale.	5	Must	US-008	S1	Data Understanding	R1
US-010	EP-03	Build a reproducible Excel ingestion layer	As a data engineer, I want repeatable Excel ingestion so that the same source extracts can recreate the analytical dataset.	- Multiple source files can be loaded through a consistent procedure. - Column names and data types are standardised. - The process records source filename and ingestion timestamp.	5	Must	US-005	S1	Data Preparation	R1
US-011	EP-03	Standardise timestamps and time zones	As an analyst, I want a consistent time representation so that time-series aggregation and external joins are correct.	- Timestamp parsing is deterministic. - Invalid timestamps are reported. - A single documented time convention is used throughout the analytical dataset.	3	Must	US-008, US-010	S1	Data Preparation	R1
US-012	EP-03	Detect irregular sampling frequencies	As a data engineer, I want sampling irregularities detected so that resampling decisions are explicit.	- Sampling intervals are calculated per relevant series. - Expected/observed frequencies are compared. - Irregular periods are flagged for treatment.	5	Must	US-008, US-011	S1	Data Preparation	R1
US-013	EP-03	Define resampling and aggregation rules	As an analyst, I want documented resampling rules so that energy and environmental signals remain analytically meaningful.	- A target modelling granularity is defined and documented. - Aggregation functions are specified by variable type. - No interpolation is applied blindly to variables for which it is inappropriate.	5	Must	US-012	S2	Data Preparation	R1
US-014	EP-03	Detect device-reset interference	As a data engineer, I want reset-related interference identified so that hardware artefacts do not distort model training.	- Reset signatures are detected using documented rules/statistics. - Affected intervals are tagged. - The original values remain recoverable for audit.	8	Must	US-009, US-012	S2	Data Preparation	R1
US-015	EP-03	Treat missing and invalid observations	As an analyst, I want missing/invalid values treated according to documented rules so that	- Each treatment is traceable to a rule. - Imputed/interpolated values are flagged.	8	Must	US-013, US-014	S2	Data Preparation	R1

ID	Epic	Story / outcome	User story	Acceptance criteria	SP	MoSCoW	Dependencies	Target	CRISP-DM	Risk
			the modelling dataset is coherent.	• Rows that cannot be reliably repaired are excluded with a documented reason.						
US-016	EP-03	Produce a clean analytical dataset	As the modelling team, we want a clean, validated dataset so that feature engineering can start from a controlled baseline.	• Schema is validated. • Temporal, missingness and reset checks pass against defined thresholds. • Data-quality summary is published. • Dataset version is recorded.	8	Must	US-015, US-022	S2	Data Preparation	R1
US-017	EP-04	Represent the Building → Floor → Zone hierarchy	As the project, we want an explicit spatial hierarchy so that energy consumption can be interpreted in physical context.	• Every mapped record can be traced to building, floor and zone where the source data permits. • Hierarchy rules are documented. • Unmapped records are flagged.	5	Must	US-007	S1	Data Understanding / Preparation	R2
US-018	EP-04	Map device IDs to zones	As an analyst, I want devices linked to their zones through unique IDs so that telemetry is spatially attributable.	• Mappings use the established unique device IDs. • Manual ad-hoc joins are not used as the authoritative mapping. • Ambiguous mappings are flagged for review.	8	Must	US-007, US-017	S2	Data Preparation	R2
US-019	EP-04	Validate orientation-based zone naming	As a data-quality owner, I want zone naming conventions validated so that identifiers such as 1NE/1NO are interpreted consistently.	• Naming patterns are documented. • Floor number and orientation components are parsed where valid. • Invalid or unexpected labels are flagged rather than silently transformed.	3	Must	US-017	S2	Data Preparation	R2
US-020	EP-04	Reconcile floor-plan context with dataset mappings	As a reviewer, I want spatial mappings checked against the supplied floor-plan visuals so that the semantic model reflects the documented building structure.	• Representative device/zone mappings are cross-checked against the supplied plans. • Exceptions are recorded. • No unsupported physical location is inferred from visual proximity alone.	5	Should	US-018, US-019	S2	Data Preparation	R2
US-021	EP-04	Handle floor-specific HVAC context	As an analyst, I want floor-specific HVAC context represented where documented so that model features do not assume identical systems across floors.	• Known floor-level HVAC information is captured where available. • The third-floor note indicating four HVAC systems is preserved as source context. • Unknown HVAC topology is not invented.	3	Should	US-020	S2	Data Preparation	R2
US-022	EP-04	Create a spatial validation report	As the team, we want a spatial QA report so that hierarchy and device-to-zone integrity are demonstrable.	• Coverage of Building → Floor → Zone mapping is quantified. • Unmapped/ambiguous devices are listed. • QA evidence is stored with the release documentation.	5	Must	US-018, US-019	S2	Data Preparation	R2
US-023	EP-05	Create energy consumption features	As an ML engineer, I want energy variables transformed into modelling features so that consumption patterns can be learned.	• Relevant energy-meter telemetry is selected from BM321/BMT01/BMP02 data where available. • Lag/rolling/calendar features are documented. • Feature generation is reproducible.	8	Must	US-016, US-022	S3	Data Preparation	R3
US-024	EP-05	Create environmental features	As an ML engineer, I want temperature, humidity, CO2 and VOC information incorporated where available so that indoor context can be evaluated.	• BSens/BSA02 variables present in the data are identified. • Temporal alignment with energy data is deterministic. • Missing environmental data is handled according to the data-quality rules.	5	Must	US-016, US-022	S3	Data Preparation	R3
US-025	EP-05	Create lighting features	As an ML engineer, I want BSL01 lighting information represented where available so that lighting-related consumption can be assessed.	• Available lighting telemetry is identified. • Aggregation aligns with the modelling time grid. • Feature usefulness can be evaluated.	3	Should	US-016, US-022	S3	Data Preparation	R3
US-026	EP-05	Create calendar features	As an analyst, I want calendar context represented so that recurring operational patterns can be captured.	• Day-of-week and time-of-day features are created where appropriate. • Academic/calendar data is considered as optional contextual enrichment. • Any external calendar source is documented.	3	Must	US-016	S3	Data Preparation	R3
US-027	EP-05	Assess Madrid weather enrichment	As an ML engineer, I want to assess Madrid weather data so that its value for energy forecasting can be evidenced rather than assumed.	• Candidate weather variables are documented. • Temporal alignment is reproducible. • Model performance with and without weather is compared using the same evaluation protocol.	5	Could	US-016, US-026	S3	Data Preparation / Modeling	R3
US-028	EP-05	Build the final modelling matrix	As the modelling team, we want a documented feature matrix so that training and evaluation use a controlled dataset.	• Target variable is explicit. • Feature list and provenance are documented. • Train/validation/test split strategy is defined without temporal leakage.	8	Must	US-023, US-024, US-026	S3	Data Preparation	R3
US-029	EP-06	Define forecasting target and evaluation protocol	As an ML engineer, I want a clear forecasting target and evaluation protocol so that model results are comparable and defensible.	• Forecast horizon/target definition is documented. • Evaluation metrics are selected and justified. • Temporal validation prevents leakage.	5	Must	US-028	S3	Modeling	R3
US-030	EP-06	Build a simple baseline model	As the modelling team, we want a transparent baseline so that later models can be evaluated	• At least one simple baseline is trained. • Baseline metrics are recorded.	5	Must	US-029	S3	Modeling	R3

ID	Epic	Story / outcome	User story	Acceptance criteria	SP	MoSCoW	Dependencies	Target	CRISP-DM	Risk
			against a known reference.	Results are reproducible from versioned data/features.						
US-031	EP-06	Train a tree-based predictive model	As an ML engineer, I want a non-linear baseline such as Random Forest so that interactions in IoT features can be captured.	- Model is trained using the approved split. - Hyperparameters are documented. - Performance is compared with the baseline.	8	Must	US-030	S4	Modeling	R3
US-032	EP-06	Perform model evaluation and error analysis	As a reviewer, I want model errors analysed so that the DataPack is evidence-based and limitations are visible.	- Performance is reported by agreed metrics. - Residual/error patterns are inspected across relevant time/spatial segments where data permits. - Potential overfitting is assessed.	8	Must	US-030, US-031	S4	Evaluation	R3
US-033	EP-06	Compare enriched vs non-enriched models	As the modelling team, we want an evidence-based comparison of contextual features so that optional enrichment is retained only when useful.	- A consistent evaluation protocol is used. - Weather/calendar additions are evaluated against a comparable baseline. - The final feature set is justified with recorded evidence.	5	Should	US-027, US-032	S4	Evaluation	R3
US-034	EP-06	Tune and select the final candidate model	As the team, we want a final candidate model selected using documented criteria so that the deliverable is coherent and reproducible.	- Candidate models are compared against the agreed metrics. - Selection rationale is recorded. - Final model artefact and parameters are versioned.	8	Must	US-032, US-033	S4	Evaluation	R3
US-035	EP-06	Document model limitations and applicability	As a stakeholder, I want limitations stated clearly so that forecasts are not interpreted beyond the evidence.	- Known data limitations are documented. - Unsupported real-time use is explicitly excluded. - Spatial/temporal coverage limitations are stated where relevant.	3	Must	US-034	S4	Evaluation	R3
US-036	EP-07	Generate spatial energy profiles	As a building-management stakeholder, I want energy results contextualised by floor and zone so that the value of the spatial model is visible.	- Results can be aggregated or analysed by floor/zone where coverage permits. - Zone labels are traceable to the data dictionary. - Charts distinguish measured history from predictions.	5	Must	US-034, US-022	S5	Evaluation	R2/R3
US-037	EP-07	Generate model performance visuals	As a reviewer, I want clear performance charts so that model behaviour can be understood during the final pitch.	- At least one actual-vs-predicted visual is produced. - Error metrics are visible and traceable. - Charts state the evaluation period and model version.	3	Must	US-034	S5	Evaluation	R3
US-038	EP-07	Generate data-quality evidence visuals	As a stakeholder, I want data-quality findings visualised so that the cleaning effort and its value are demonstrable.	- Missingness, sampling irregularity and reset-related cleaning are evidenced. - Visuals distinguish raw issues from cleaned results. - No chart hides a material limitation.	3	Must	US-016, US-014	S5	Data Understanding / Preparation	R1
US-039	EP-07	Build the DataPack narrative	As the project team, we want a coherent DataPack narrative so that findings directly support the project's business message.	- Problem → data → preparation → modelling → evidence → limitations → implications is coherent. - Claims are traceable to analysis outputs. - No unsupported business benefit is presented as achieved.	5	Must	US-036, US-037, US-038	S5	Deployment / Communication	R4
US-040	EP-07	Prepare the 20-minute final presentation	As the team, we want a timed presentation so that the 30 November pitch communicates the project within the required format.	- Presentation is rehearsed to approximately 20 minutes. - Required technical evidence and business message are included. - Likely Q&A topics and limitations are prepared.	8	Must	US-039	S5	Deployment / Communication	R4
US-041	EP-07	Prepare the 20-minute Q&A evidence pack	As the team, we want an evidence pack so that questions can be answered from traceable project facts.	- Data sources, methodology, model metrics, spatial mapping and limitations can be located quickly. - Key decisions are documented. - Unsupported assumptions are clearly identified.	5	Must	US-040	S5	Deployment / Communication	R4
US-042	EP-08	Implement automated data-quality checks	As QA, I want automated checks so that regressions in the analytical dataset are detected early.	- Schema, null, timestamp and key integrity checks run automatically. - Failures are reported clearly. - Checks can be rerun after data refreshes.	5	Must	US-016	S2-S4	All	R1
US-043	EP-08	Implement reproducible modelling pipeline	As an ML engineer, I want the modelling workflow reproducible so that results can be regenerated for review.	- Data version, feature version, model parameters and output artefacts are linked. - A clean environment/setup instruction exists. - A second team member can reproduce the final result using the documented process.	8	Must	US-028, US-034	S4	Modeling / Evaluation	R3
US-044	EP-08	Perform model validation review	As a model-validation owner, I want an independent review of the final model evidence so that acceptance is controlled.	- Evaluation protocol is checked. - Potential leakage/overfitting is reviewed. - Acceptance findings and residual issues are recorded.	5	Must	US-034, US-043	S4	Evaluation	R3
US-045	EP-08	Review scope and out-of-scope compliance	As the PM, I want a scope compliance review so that no excluded capability is accidentally delivered as part of the project.	- No hardware integration work is included. - No live pipeline modification is included. - No real-time streaming architecture is included. - Historical Excel scope is respected.	3	Must	US-039	S5	All	R4
US-046	EP-08	Review academic/internal data-use constraints	As the project team, I want documented compliance with the stated UPM educational/internal research basis so that data use remains within the project	- Data use is documented as educational/internal research per the Business Case. - No PII is intentionally introduced into the modelling dataset. - External datasets, if used, have their source and usage basis	3	Must	US-028	S4	All	R5

ID	Epic	Story / outcome	User story	Acceptance criteria	SP	MoSCoW	Dependencies	Target	CRISP-DM	Risk
			assumptions.	documented.						
US-047	EP-09	Freeze final dataset and model versions	As the release team, we want final artefacts frozen so that the presentation and handover refer to stable results.	- Final dataset/feature/model versions are tagged. - Presentation metrics are generated from the frozen versions. - Any post-freeze change is recorded.	3	Must	US-044, US-045	S5	Deployment	R4
US-048	EP-09	Package source code and documentation	As the client, I want a complete code/documentation package so that the work can be reviewed and reused.	- Source code, notebooks/scripts, README/setup instructions, data dictionary and model documentation are included. - No prohibited raw data is bundled. - Package structure is documented.	5	Must	US-043, US-047	S5	Deployment	R4
US-049	EP-09	Prepare final Moodle handover	As the project owner, I want the final release prepared for Moodle so that the 14 December deadline is met.	- All required code and documentation are included. - Release checklist is signed off internally. - Upload package is verified for completeness.	3	Must	US-048	S5	Deployment	R4
US-050	EP-09	Conduct final acceptance and lessons learned	As the project stakeholders, we want a formal close-out so that acceptance, remaining limitations and lessons learned are captured.	- Final presentation outcome is recorded. - Acceptance/sign-off status is documented. - Lessons learned and backlog carry-over items are recorded. - Project closure evidence is archived.	5	Must	US-040, US-049	Closure	Deployment / Communication	R4

4. Epic-Level Acceptance and Deliverables

EP-01 - Project setup, governance & backlog control

A governed project baseline exists and the team can demonstrate controlled prioritisation, change management and sprint acceptance.

Expected deliverables: Repository and collaboration structure; Baselined backlog; Sprint cadence; Decision/change/risk traceability.

EP-02 - Data inventory & data dictionary

Every relevant source and telemetry concept is catalogued sufficiently to support controlled preparation and modelling.

Expected deliverables: Source inventory; Telemetry catalogue; Central data dictionary; Temporal coverage and quality profile.

EP-03 - Data ingestion & quality engineering

Historical Excel extracts can be ingested reproducibly and transformed into a quality-controlled analytical dataset.

Expected deliverables: Reproducible ingestion; Timestamp normalisation; Sampling treatment; Reset interference handling; Missing-value treatment; Clean dataset.

EP-04 - Spatial modelling: building → floor → zone

Analytical records are traceable to the documented physical hierarchy and device IDs without unsupported manual assumptions.

Expected deliverables: Spatial hierarchy; Device-to-zone mapping; Zone naming validation; Plan cross-check; Spatial QA report.

EP-05 - Feature engineering & contextual enrichment

A modelling matrix contains traceable energy, environmental, lighting and calendar/context features as supported by the data.

Expected deliverables: Energy features; Environmental features; Lighting features; Calendar features; Optional weather assessment; Final modelling matrix.

EP-06 - Predictive modelling & evaluation

At least one functional predictive energy model is evaluated using a leakage-safe, documented protocol.

Expected deliverables: Forecast target; Baseline; Tree-based model; Error analysis; Enrichment comparison; Final candidate; Limitations.

EP-07 - Insights, visualisation & communication

The DataPack communicates evidence, model performance, spatial insights and limitations within the required final presentation format.

Expected deliverables: Spatial profiles; Performance visuals; Data-quality evidence; Narrative; 20-minute pitch; Q&A evidence.

EP-08 - Quality, risk, security & reproducibility

The solution is testable, reproducible and demonstrably compliant with scope and stated data-use assumptions.

Expected deliverables: Automated QA; Reproducible pipeline; Model validation; Scope compliance; Data-use review.

EP-09 - Final release, handover & closure

Stable code, model documentation and evidence are packaged and handed over by the required deadlines.

Expected deliverables: Version freeze; Release package; Moodle handover; Acceptance and lessons learned.

5. Sprint Plan and Release Roadmap

The roadmap follows the Business Case: Sprint 0 for setup/backlog; Sprints 1–2 for data engineering; Sprints 3–4 for predictive modelling; Sprint 5 for finalisation and pitch preparation. The backlog should be re-planned at each Sprint Planning event based on actual data findings, capacity and risk.

Sprint	Window	Theme	Key stories	Methodology alignment	Exit outcome
S0	Mid-Sep	Setup & backlog	US-001–004	PM ² initiation/planning; Business Understanding	Baselined backlog, repository, governance cadence
S1	Late Sep–early Oct	Data understanding & ingestion	US-005–012	CRISP-DM Data Understanding / Preparation	Inventory, dictionary, profiling, ingestion, timestamps
S2	Early–mid Oct	Data quality & spatial model	US-013–022, US-042	CRISP-DM Data Preparation	Clean dataset, reset treatment, Building→Floor→Zone mapping
S3	Late Oct	Feature engineering & baseline modelling	US-023–030	CRISP-DM Data Preparation / Modeling	Feature matrix, forecast target, baseline
S4	Early–mid Nov	Model evaluation & validation	US-031–035, US-043–046	CRISP-DM Modeling / Evaluation	Candidate model, comparison, validation, limitations
S5	Late Nov	Insights, presentation & release	US-036–041, US-047–050	CRISP-DM Deployment / Communication; PM ² Closing preparation	DataPack narrative, 20-min pitch, Q&A pack, release

Hard milestones

30 November 2026: final 20-minute presentation + 20-minute discussion. 14 December 2026: final release of all materials, code and documentation through Moodle.

6. Traceability Matrix

This matrix connects the backlog to the Business Case objectives, explicit scope, success criteria and principal risks.

Trace ID	Business Case area	Backlog coverage	Evidence / interpretation	Risk
BC-O1	Data preparation	US-005–022, US-042	Clean and structure historical IoT extracts; handle sampling irregularities, reset noise and spatial mapping.	R1/R2
BC-O2	Predictive modelling	US-023–035, US-043–044	Build/evaluate predictive energy models using CRISP-DM.	R3
BC-O3	Communication / delivery	US-036–041, US-047–050	Demonstrate evidence and deliver presentation/handover.	R4
SC-1	Successful ingestion and cleaning	US-010–016, US-042	Excel ingestion, sampling treatment and reset handling evidenced.	R1
SC-2	At least one functional predictive model	US-029–034	Model trained, evaluated and documented using spatially contextualised data.	R3
SC-3	Final presentation on 30 Nov	US-039–041	Pitch and Q&A evidence prepared.	R4
OOS-1	Hardware / infrastructure	US-045	Explicitly excluded from backlog.	R4
OOS-2	Live pipeline engineering	US-045	Explicitly excluded from backlog.	R4
OOS-3	Real-time streaming	US-045	Explicitly excluded from backlog.	R4

7. Dependency and Critical-Path Logic

- US-005 → US-006 → US-007 → US-018 is the core semantic-data dependency chain.
- US-008/009/012/014/015 → US-016 establishes the data-quality gate before modelling.
- US-017/018/019/022 establishes spatial integrity before spatially contextualised modelling and reporting.
- US-028 → US-029 → US-030/031 → US-032 → US-034 is the core modelling/evaluation chain.
- US-034/043/044 → US-047 → US-048/049 is the release chain.
- US-039 → US-040 → US-041 supports the hard 30 November presentation milestone.

7.1 Release gates

Gate	Timing	Evidence	Exit criterion
Gate G1 - Data Readiness	End S2	US-016 + US-022 + US-042	No critical unresolved data-quality or spatial-integrity issue blocks feature engineering.
Gate G2 - Model Readiness	End S4	US-034 + US-044	At least one functional predictive model is evaluated with documented limitations and reproducibility evidence.
Gate G3 - Presentation Readiness	Before 30 Nov	US-039–041	Pitch, charts, evidence pack and timing are reviewed.
Gate G4 - Handover Readiness	Before 14 Dec	US-047–050	Final artefacts are frozen, packaged, verified and submitted; closure evidence is recorded.

8. Risk-Driven Backlog Controls

ID	Risk	Assessment	Backlog controls	Mitigation	Control point
R1	Data Quality & Preprocessing Bottlenecks	High / High	US-008–016, US-042	Timebox EDA/cleaning; automate checks; preserve raw values; document imputation/interpolation.	Data Readiness Gate
R2	Spatial Context Misalignment	Medium / High	US-007, US-017–022	Central data dictionary; unique device IDs; visual cross-check; explicit ambiguity log.	Spatial QA Gate
R3	Suboptimal Model Performance	Medium / High	US-029–035, US-043–044	Start simple; evaluate systematically; enrich with contextual data only if evidence supports it; document limitations.	Model Readiness Gate
R4	Schedule Slippage & Scope Creep	Medium / Critical	US-002–004, US-039–050	Strict sprint cadence; prioritised backlog; enforce exclusions; protect hard milestones.	Presentation / Handover Gates
R5	Data-use / governance assumptions	Project assumption	US-046	Keep educational/internal research use documented; do not introduce PII; document external sources.	Release review

9. Prioritisation Rules

MoSCoW is used to maintain delivery focus under the fixed deadline. The categories are operational priorities for this release and should be reviewed during backlog refinement.

Priority	Definition in this project
Must	Required to satisfy the approved scope, a stated success criterion, a hard milestone, a key data-quality gate or a necessary release dependency.
Should	Important for quality, explainability or value, but the core release can still be completed if it is deferred without violating the approved success criteria.
Could	Useful enhancement whose inclusion depends on evidence, time and capacity; optional external enrichment is the clearest example.
Won't (this release)	Explicitly excluded or intentionally deferred. Hardware integration, live pipeline changes and real-time streaming belong here unless a formally approved scope change occurs.

10. Estimation and Capacity Model

The baseline contains 248 story points: 227 Must, 16 Should and 5 Could points. The Business Case estimates approximately six hours per team member per week across five Agile sprints. Story points are relative sizing and should not be converted directly into hours without the team's own historical velocity.

Capacity warning

Because the Business Case identifies data quality and schedule as major risks, Sprint 1–2 capacity should be protected for discovery and cleaning. If actual data complexity exceeds the baseline, preserve the Must-path to a defensible model and presentation rather than adding excluded scope.

11. Product-Level Acceptance Criteria

- Historical Excel extracts in the agreed project scope can be ingested reproducibly.
- Known irregular sampling and device-reset interference are explicitly detected and treated using documented rules.
- Data can be traced through Building → Floor → Zone and unique device IDs wherever the source supports that mapping.
- A central data dictionary exists and is used by downstream processing.
- At least one fully functional predictive model for energy consumption is trained and evaluated using the spatially contextualised dataset.
- Model evaluation uses a documented protocol and avoids temporal leakage.
- Relevant environmental, lighting and calendar features are considered where supported by the data; external data is optional and evidence-driven.
- Data-quality and model-performance evidence is documented sufficiently for review and presentation.
- The final presentation is prepared for 30 November 2026 and the complete code/documentation package is prepared for the 14 December 2026 Moodle handover.
- No excluded hardware, live-pipeline or real-time-streaming capability is delivered as part of this baseline.

12. Glossary and Controlled Terminology

Term	Controlled meaning
DataPack	The project solution/package containing the processed data workflow, predictive modelling evidence and communication outputs.
CeDInt	The instrumented UPM building used as the smart-building case.
CRISP-DM	The data-mining lifecycle used in the Business Case: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation and Deployment/communication.
Building → Floor → Zone	The spatial hierarchy used to contextualise telemetry.
Zone identifier	An orientation/floor-based identifier such as the examples 1NE and 1NO shown in the supplied material.
BM321 / BMT01 / BMP02	Energy-meter device families named in the Business Case.
BSens / BSA02	Indoor environmental sensor families named in the Business Case for temperature, humidity, CO2 and VOC-related logs.
BSL01	Smart-lighting node family named in the Business Case.
PM ²	The project-management governance framework used here to structure governance, planning, control, quality, risks and closure.
Scrum	The Agile execution framework selected in the Business Case, with two-week sprints and weekly meetings.
MoSCoW	Priority classification used for backlog management: Must, Should, Could, Won't.
DoR / DoD	Definition of Ready / Definition of Done, used to control backlog quality and completion.

13. Source Reference

Primary source used for the backlog: Business Case: Project “DataPack” – CeDInt Smart Building, version 1.2, September 2026, including the project scope, objectives, success criteria, risks, roadmap and governance. The four supplied screenshots/floor-plan visuals were used as contextual evidence for the building/floor/zone and sensor-distribution portions of the backlog.

Important source limitation

The supplied material does not provide a complete machine-readable list of every physical zone, every sensor-to-zone mapping, every telemetry unit, or a guaranteed model target horizon. Accordingly, the backlog makes these items explicit as discovery/validation work rather than inventing missing facts.

14. Backlog Baseline Checklist

- ☐ Sponsor / Product Owner validates scope and priorities.
- ☐ PM² project governance documents reference this backlog baseline.
- ☐ All source files are inventoried before modelling starts.
- ☐ Central data dictionary is approved for use.
- ☐ Spatial mapping is validated against the available evidence.
- ☐ Data readiness gate is passed before final model training.
- ☐ Model evaluation and validation evidence are approved.
- ☐ Presentation is rehearsed and timed before 30 November.
- ☐ Final artefacts are frozen and packaged before Moodle handover.
- ☐ Acceptance, lessons learned and closure evidence are recorded.